

Xplique

A Deep Learning Explainability Toolbox

Thomas Fel^{1,3,4*} Lucas Hervier^{2*}

David Vigouroux² Antonin Poche² Justin Plakoo² Remi Cadene³ Mathieu Chalvidal^{1,3}
Julien Colin^{1,3} Thibaut Boissin^{1,2} Louis Bethune¹ Agustin Picard^{5,1} Claire Nicodeme⁴
Laurent Gardes⁴ Gregory Flandin^{1,2} Thomas Serre^{1,3}

¹Artificial and Natural Intelligence Toulouse Institute, Université de Toulouse, France

²Institut de Recherche Technologique Saint-Exupéry, France

³Carney Institute for Brain Science, Brown University, USA

⁴Innovation & Research Division, SNCF , ⁵Scalian

Abstract

*Today's most advanced machine-learning models are hardly scrutable. The key challenge for explainability methods is to help assisting researchers in opening up these black boxes — by revealing the strategy that led to a given decision, by characterizing their internal states or by studying the underlying data representation. To address this challenge, we have developed **Xplique**: a software library for explainability which includes representative explainability methods as well as associated evaluation metrics. It interfaces with one of the most popular learning libraries: Tensorflow as well as other libraries including PyTorch, scikit-learn and Theano. The code is licensed under the MIT license and is freely available at github.com/deel-ai/xplique.*

1. Introduction

Deep neural networks [27, 40] are widely used in many applications including medicine, transportation, security and finance, with broad societal implications [5, 23, 34]. Yet, these networks have become almost impenetrable. Furthermore, in most real-world scenarios, these systems are used to make critical decisions, often without any explanation. A growing body of research thus focuses on making those systems more trustworthy via the development of explainability methods to make their predictions more interpretable [8]. Such methods will find broad societal uses and will help to fulfill the “right to explanation” that European laws guarantee to its citizens [21]. Hence, it is important for explainability methods to be made widely available. In-

deed, several libraries have already been proposed including Captum [25] for Pytorch.

In this work, we propose the first of such libraries – based on Tensorflow [1]. Our library includes all main explainability approaches including: (1) attribution methods (and their associated metrics), (2) feature visualization methods and (3) concept-based methods.

1.1. Attribution methods

aim to produce so-called saliency maps or more simply, heatmaps, to explain models' decisions. These maps reveal the discriminating input variables used by the system for arriving to a given decision. The score assigned to a region of an image (or a word in a sentence) reflects its importance for the prediction of the model. We have reimplemented more than 14 representative explanation methods [2, 7, 9, 11, 13, 29, 33, 35, 38, 39, 41, 43, 44, 48–50]. We provide support for images, tabular data and time series. As one can imagine, the large number of explanation methods available has brought to the forefront a major issue: the urgent need for metrics to evaluate explanations. Indeed, inconsistencies produced across these methods have raised questions about their legitimacy [2–4, 6, 10, 12, 16, 17, 19, 20, 26, 28, 36, 42, 45, 46]. Our implementation thus also includes several common metrics associated with these attribution methods.

1.2. Feature Visualization

Even though attribution methods are sometimes useful to understand a decision, they leave aside the global study of a Deep Learning model. Several methods attempt to tackle this issue including feature visualization methods for studying the internal representations learned by a model.

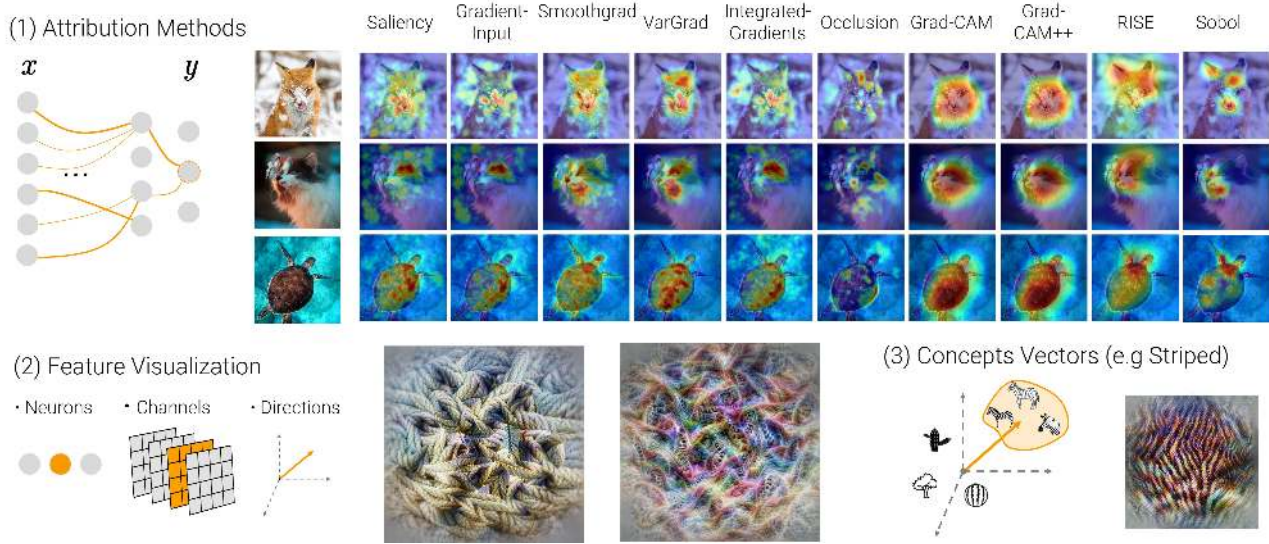


Figure 1. **Xplique modules.** The library contains 3 main modules: (1) an “Attribution Methods” module, (2) a “Feature Visualization” module and (3) a “Concepts” module .

The method proposed in [30–32] is a popular technique employed to explain the internal representations of a model. This method aims to find an interpretable input (or stimulus) that maximizes the response of a given neuron, a set of neurons (e.g., a channel) or a direction in an internal space of the model. Thus, the corresponding stimulus is a prototype of what the neuron responds to. We provide an API able to optimize such input by targeting a layer, a channel, a direction or combinations of these objectives. The optimization tool leverages the latest advances in the field (e.g., Fourier preconditioning, robustness to transformations).

1.3. Concept-based methods

Nevertheless, the interpretation of feature visualization methods is left to the user. Fortunately, another approach consists in letting the user derive concept vectors that are meaningful to them: Concept-based methods.

[14, 15, 18, 22, 24, 37, 47] work on high-level features interpretable by humans. This includes a method to retrieve Vectors of Activations of these human Concepts (CAV) [22]. These vectors help to make the passage between human concepts and a vector base formed by the neurons of a model at a specific layer. In addition, we have also re-implemented TCAV, which then tests how important these human vectors are to the model’s decisions.

Finally, the library also allows interactions between all 3 modules such that one can leverage the feature visualization module to visualize the extracted CAV (see Fig.1) or the feature attribution module to visualize the location of the CAV on an image. A major effort has been made to facilitate the use of the software and various examples are provided as notebooks for each of the modules.

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¹<https://www.deel.ai/>

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